SKILL 2

Data Preprocessing of the One Piece Bounty Dataset

Reformatted Lab Report • Cleaning • Feature Preparation • Outlier Flagging

Raw dataset	165 records and 3 source columns
Prepared data	165 records, with 145 known bounties and 20 unknown bounties retained
Main tools	Python, pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

1. Aim

To standardize and clean the One Piece bounty dataset, convert bounty text into numeric values, explicitly preserve unknown values, rebuild ranks, create scaled features, and flag potential outliers.

2. Procedure

1. Load the raw CSV and inspect its structure.
2. Standardize column headers and clean text fields.
3. Check for duplicate names.
4. Parse bounty strings and correct the currency symbol inconsistency.
5. Keep unknown bounties as missing values instead of imputing them.
6. Rebuild ranks and derive billion-scale and log10 bounty features.
7. Flag outliers with IQR fences on the log scale and validate the results.

3. Observation

The input file contains 165 rows and 3 columns. The cleaning process standardized the headers, converted 145 bounty strings to numeric values, corrected one inconsistent currency symbol, and kept 20 unknown bounty values unfilled. Placeholder ranks were nulled and ranks were reconstructed from bounty values. The highest known bounty belongs to Gol D. Roger at 5,564,800,000 Berries. Raw bounty skewness was 4.28. On the log10 scale, skewness was -2.44; after excluding the two flagged low-value outliers, it was approximately 0.03. Bepo and Tony Tony Chopper were flagged as outliers but retained in the data.

4. Result and Conclusion

The preprocessing workflow generated a structured dataset with explicit missing-value flags, corrected currency formatting, rebuilt ranks, scaled features, and outlier indicators. All 165 records were retained. The cleaned dataset can be used for later exploratory analysis and feature engineering.

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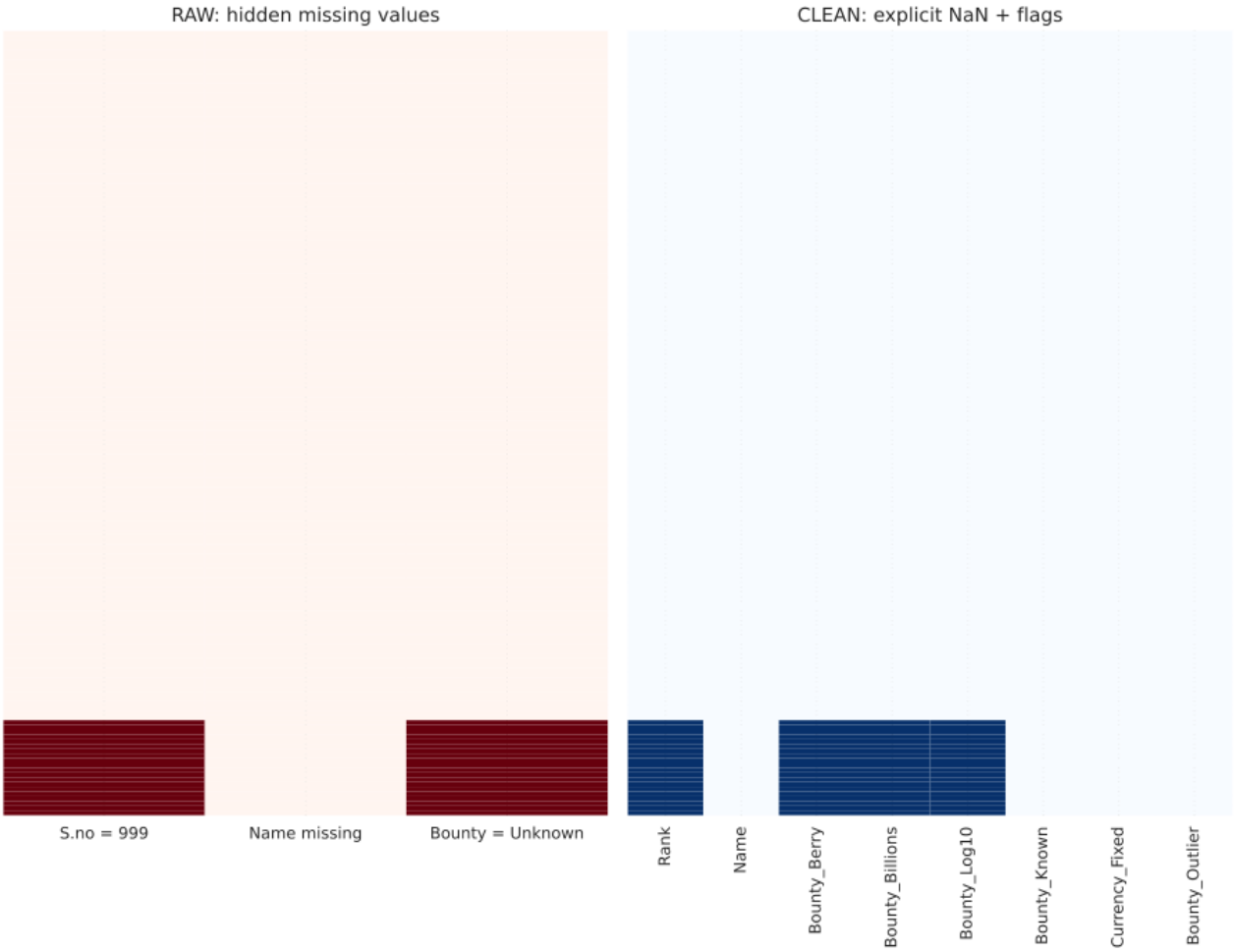
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SKILL 2 - PREPROCESSING REPORT
One Piece Bounty Dataset

Raw shape   : (165, 3)
Clean shape : (165, 8)
Known bounties   : 145
Unknown bounties : 20

Issues resolved
  - Headers standardised: 3
  - Names tidied: 0
  - Duplicate pirates removed: 0
  - Bounty strings parsed to numbers: 145
  - Wrong currency symbols fixed: 1
  - Unknown bounties flagged (not imputed): 20
  - Placeholder ranks (999) nulled: 20
  - Rank mismatches rebuilt: 0
  - Outliers flagged (log IQR): 2

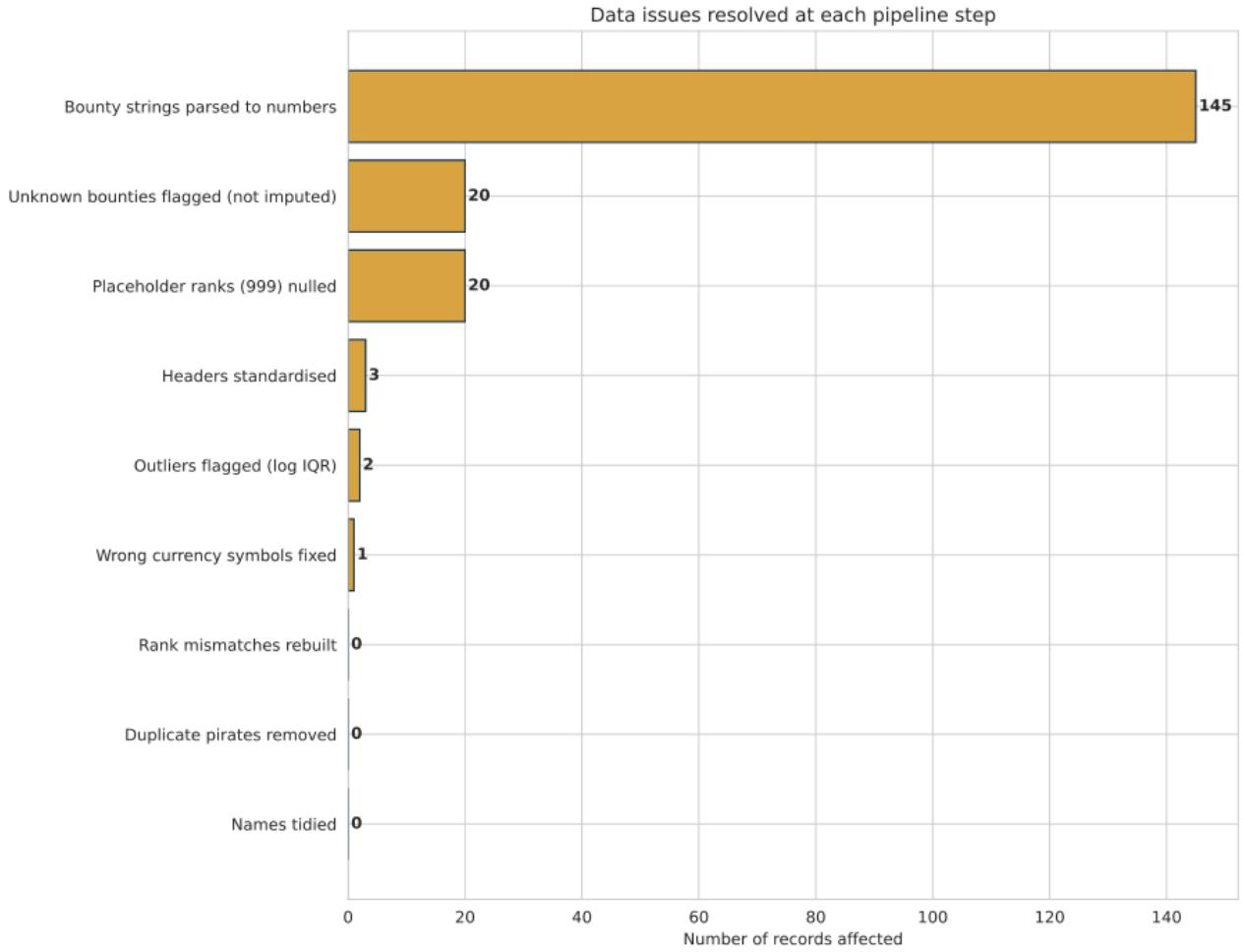
Skewness before log : 4.28
Skewness after log  : -2.44
After log, minus outliers : 0.03
```

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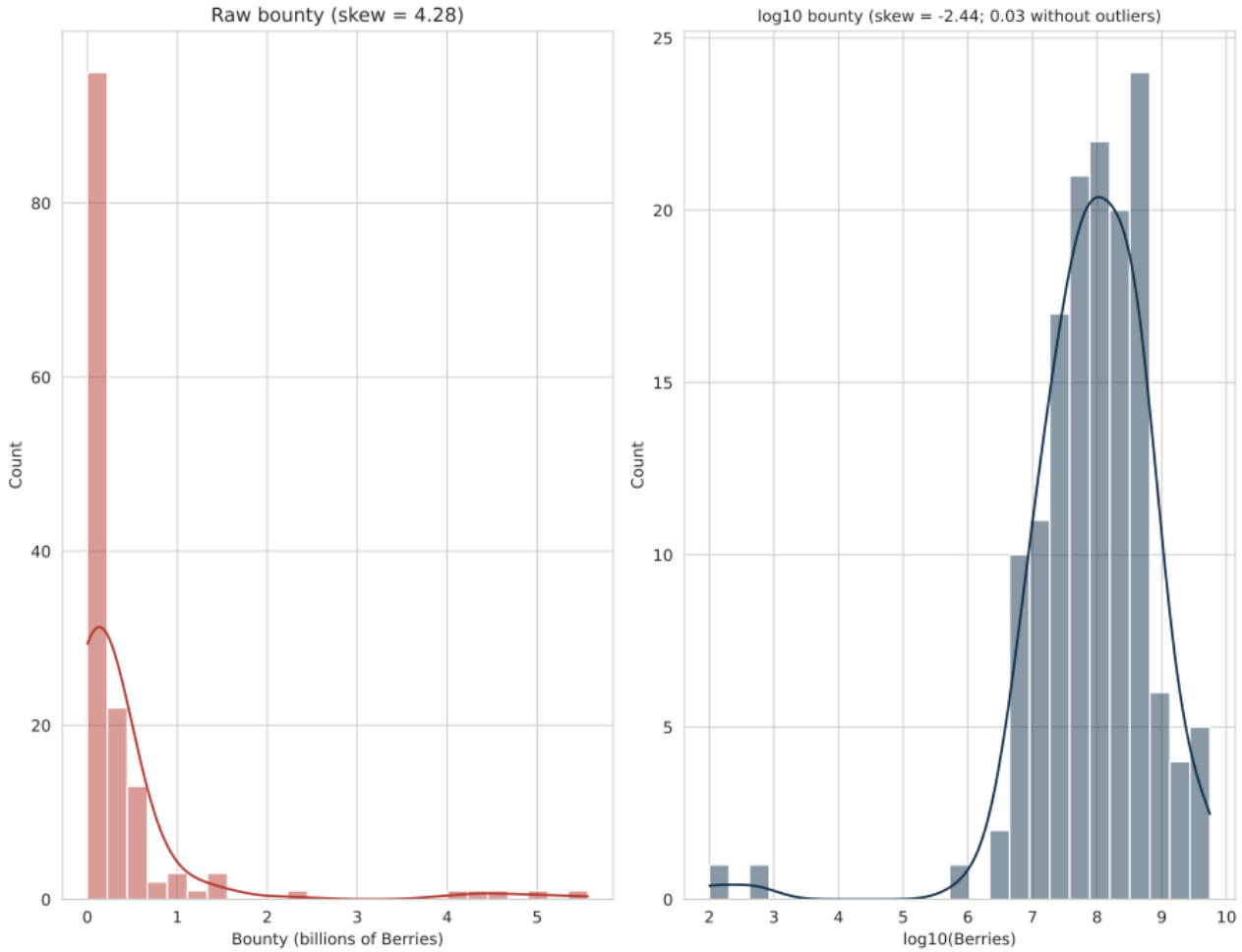
The source figure compares raw hidden missing values with explicit missing-value indicators in the cleaned dataset.

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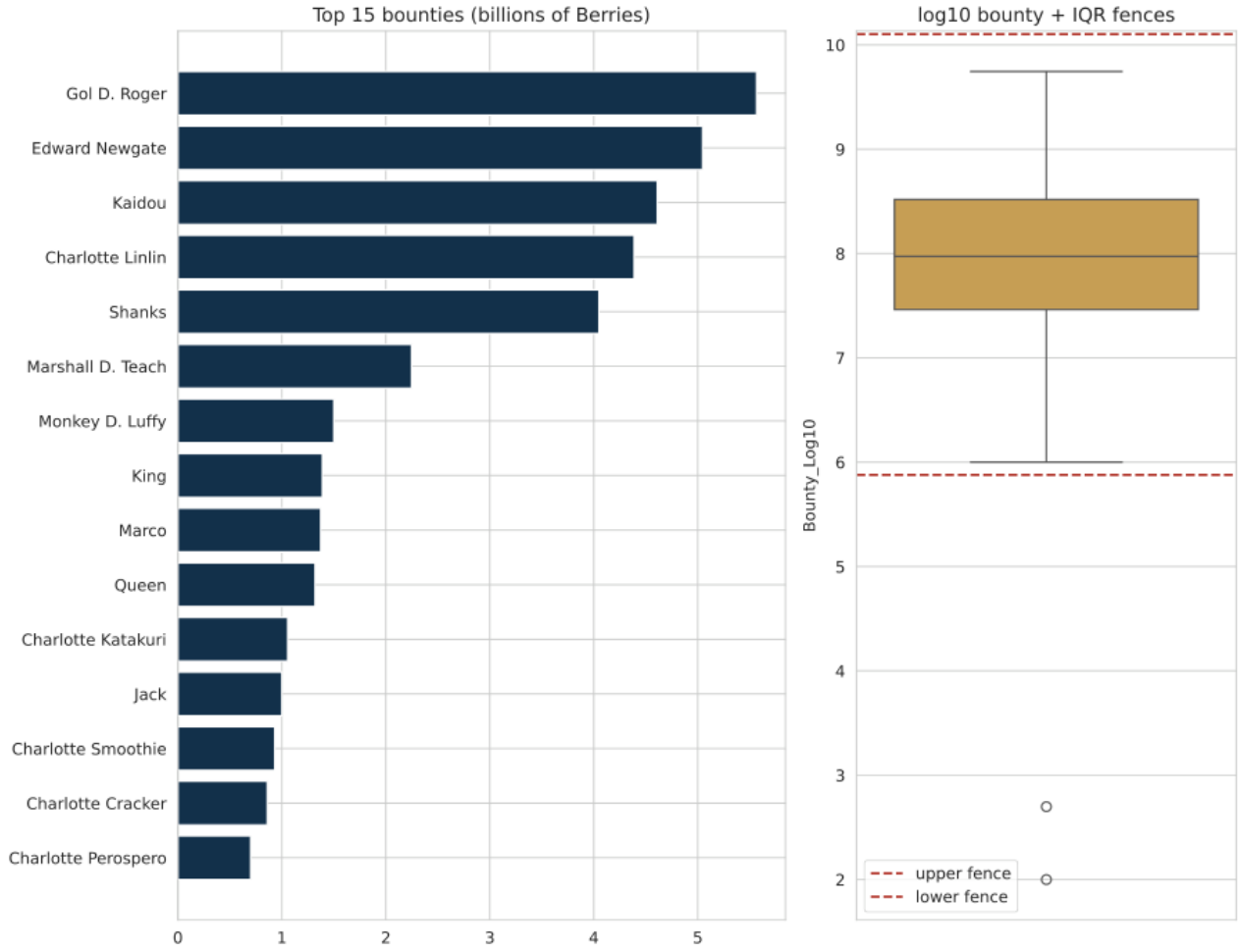
The pipeline chart summarizes the number of records affected by each preprocessing step.

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The histograms compare raw bounty distribution with the log10-transformed distribution.

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The fifth source page shows the 15 largest known bounties and the IQR fences used for outlier review.

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PIPELINE AUDIT TRAIL

01. Load raw data              rows=165  nulls=0
02. Standardise headers        rows=165  nulls=0
03. Clean text fields          rows=165  nulls=0
04. Remove duplicate names     rows=165  nulls=0
05. Parse bounty strings       rows=165  nulls=20
06. Currency sanity check      rows=165  nulls=20
07. Flag unknown bounties      rows=165  nulls=20
08. Rebuild rank               rows=165  nulls=40
09. Sort + scale bounty        rows=165  nulls=80
10. Flag outliers              rows=165  nulls=80
11. Validation passed          rows=165  nulls=80
```